

Manit Hub

Database Schema Reference — MongoDB Collections (Mongoose Models)

21 collections

MongoDB + Mongoose

Multi-tenant by University

Generated July 2026

Source: `backend/models/*.js` · Campus super-app for MANIT students — marketplace, notes, Q&A, chat, study groups, rides, events, attendance & more.

Overview — All Collections

Conventions used throughout: every collection has an implicit `_id` (ObjectId) primary key and `createdAt` / `updatedAt` timestamps (`timestamps: true`). Almost every collection carries a required, indexed `university` → `University` reference — the app is multi-tenant and every query is scoped to the user's university. * marks required fields. `ObjectId→X` means a reference to collection X.

MODEL	COLLECTION	DOMAIN	PURPOSE
User	users	Identity	Student accounts, auth, profile, gamification, moderation state
University	universities	Identity	Tenant root — campus + allowed email domains
Friendship	friendships	Social	Friend requests / accepted friendships (one doc per user pair)
Listing	listings	Marketplace	Items for sale (textbooks, electronics, ...)
Offer	offers	Marketplace	Price offers & counter-offers on listings
Conversation	conversations	Chat	1-to-1 chat threads (listing / lost-found / ride / friend DM)
Message	messages	Chat	Individual chat messages with read receipts
Document	documents	Academics	Shared study material (notes, PYQs, syllabus) with votes & comments
Question	questions	Academics	Q&A forum questions
Answer	answers	Academics	Answers to questions, upvotable, acceptable
StudyGroup	studygroups	Academics	Study groups with members, links & scheduled sessions
AcademicRecord	academicrecords	Academics	Per-semester grades for CGPA calculation
AttendanceSubject	attendancesubjects	Academics	Per-subject attendance tracker (attended / held / target%)
TimetableEntry	timetableentries	Academics	Weekly class schedule slots with reminders
Confession	confessions	Campus life	Anonymous confessions with reactions & comments
Event	events	Campus life	Campus events with attendees & reminders
LostFoundItem	lostfounditems	Campus life	Lost & found posts
Ride	rides	Campus life	Ride-sharing posts with seats & passengers
Notification	notifications	Infra	In-app notification feed (polymorphic related entity)
DeviceToken	devicetokens	Infra	Push-notification device tokens (web/android/ios)
Report	reports	Moderation	User reports on any content type, with admin review queue

1 · Identity & Tenancy

User

users

Student accounts: authentication, profile, wallet/UPI, preferences, gamification and moderation state.

FIELD	TYPE	CONSTRAINTS / NOTES
email *	String	unique, lowercase; must match <code>^\d{10}@stu.manit.ac.in\$</code> (MANIT scholar email)
password *	String	min 8 chars; <code>select:false</code> ; bcrypt-hashed in a pre-save hook
displayName *	String	trimmed
handle	String	unique + sparse @handle (Instagram-style), <code>^[a-z0-9_.]{3,20}\$</code>
phone / bio / location	String	optional profile fields, default ""
avatarUrl	String	Cloudinary URL, default ""
university *	ObjectId→University	indexed — tenant scope
isAdmin	Boolean	default false; grants moderation queue access
points	Number	default 0, indexed — gamification (leaderboard)
badges	[String]	earned badge slugs
savedListings	[ObjectId→Listing]	wishlist
paymentInfo.upild	String	validated UPI pattern <code>^[w.\-]{2,}@[a-zA-Z]{2,}\$</code>
paymentInfo.upiQrUrl	String	uploaded QR image URL
notificationPreferences	Object	messages / studyGroups / marketplace — Booleans, default true
privacySettings.profileVisibility	String enum	everyone students private (default everyone)
privacySettings.showOnlineStatus	Boolean	default true
privacySettings.allowDirectMessages	Boolean	default true
isActive	Boolean	default true (soft delete)
isBanned / strikes / banReason / bannedAt	Boolean / Number / String / Date	moderation: strikes accumulate on removed content; ban blocks login, API & socket
passwordResetToken / passwordResetExpires	String / Date	<code>select:false</code> — forgot-password flow
tokenVersion	Number	<code>select:false</code> ; bumped on password reset to invalidate all issued JWTs
emailVerified	Boolean	default true (legacy accounts stay valid); only accounts created with <code>false</code> are gated
emailVerificationToken / emailVerificationExpires	String / Date	<code>select:false</code> — email ownership verification

Hooks & methods:

- pre-save: `bcrypt-hash password` (cost 10) when modified
- `comparePassword(candidate)` instance method for login

University

universities

Tenant root. Every other collection references it; email domains gate signup.

FIELD	TYPE	CONSTRAINTS / NOTES
name	String	display name
domains *	[String]	unique — allowed student email domains
isVerified	Boolean	default false
isActive	Boolean	default true

Friendship

friendships

One document per pair of users — friend request lifecycle (pending → accepted).

FIELD	TYPE	CONSTRAINTS / NOTES
requester *	ObjectId→User	indexed — who sent the request
recipient *	ObjectId→User	indexed — who received it
users	[ObjectId→User]	exactly 2, kept sorted (pre-validate hook)
pairKey	String	unique — canonical "<lowId>_<highId>"; scalar because a unique index on the array would be multikey (one friendship per user!)
status	String enum	pending accepted (default pending, indexed)
university *	ObjectId→University	indexed

Indexes: {pairKey:1} unique · {users:1, status:1} **Hooks:** pre-validate sorts users and derives pairKey

2 · Marketplace

Listing

listings

Items students sell to each other.

FIELD	TYPE	CONSTRAINTS / NOTES
title *	String	trimmed; part of text index
description	String	part of text index
price *	Number	min 0
category *	String enum	Textbooks Electronics Furniture Clothing Sports Other
images	[String]	Cloudinary URLs
seller *	ObjectId→User	
university *	ObjectId→University	indexed
condition	String enum	new like-new good fair (default good)
status	String enum	available sold reserved (default available, indexed)
isActive	Boolean	default true (soft delete)

Indexes: text {title, description} · {university:1, isActive:1}

Offer

offers

Haggling: buyers make offers, sellers accept / decline / counter.

FIELD	TYPE	CONSTRAINTS / NOTES
listing *	ObjectId→Listing	indexed
buyer *	ObjectId→User	indexed
seller *	ObjectId→User	indexed; denormalized from the listing for fast "offers received" queries
university *	ObjectId→University	indexed
amount *	Number	min 1
message	String	max 300 chars
counterAmount	Number	min 1 — seller's counter (status becomes "countered")
status	String enum	pending countered accepted declined withdrawn (default pending, indexed)

Indexes: {listing:1, buyer:1, status:1}

3 · Chat & Messaging

Conversation

conversations

1-to-1 chat threads. Polymorphic: started from a listing, a lost-&-found item, a ride, or as a friend DM.

FIELD	TYPE	CONSTRAINTS / NOTES
listingId	ObjectId→Listing	polymorphic context id — Listing, LostFoundItem or Ride depending on contextType ; null for friend DMs
contextType	String enum	listing lostfound ride friend (default listing)
listingTitle	String	denormalized title; empty for friend DMs (UI shows the other participant)
participants *	[ObjectId→User]	exactly 2; sorted in pre-save for canonical ordering
university *	ObjectId→University	indexed
lastMessage	String	denormalized preview for the inbox list
lastMessageAt	Date	default now, indexed — inbox sort key

Indexes: {participants:1} (non-unique lookup index — a unique multikey index here wrongly caps each user at one conversation per listing; duplicates are prevented in the application layer instead)

Message

messages

Individual chat messages, delivered live over Socket.IO.

FIELD	TYPE	CONSTRAINTS / NOTES
conversation *	ObjectId→Conversation	indexed
sender *	ObjectId→User	indexed
university *	ObjectId→University	indexed
content *	String	trimmed
readAt	Date	null until read — read receipts
type	String enum	text system (default text)
isDeleted	Boolean	default false (soft delete)

Indexes: {conversation:1, createdAt:1} — chat history loading

4 · Academics & Study

Document

documents

Shared study material — notes, previous-year questions, syllabus — with voting and comments.

FIELD	TYPE	CONSTRAINTS / NOTES
title *	String	part of text index
description	String	
type *	String enum	Notes PYQ Syllabus Sem Schedule Other
branch / subject / semester	String	filters; subject is in the text index
year	Number	
fileUrl *	String	Cloudinary URL of the file
fileName / fileFormat	String	original name and extension
fileSize	Number	bytes
uploader *	ObjectId→User	
university *	ObjectId→University	indexed
downloadCount	Number	default 0
upvotes	[ObjectId→User]	voter list (one vote per user)
upvoteCount	Number	default 0, indexed — denormalized for "Top rated" sort
comments	[Subdoc]	{author* → User, content* (max 500), createdAt}
isActive	Boolean	default true

Indexes: text {title, subject} · {university:1, isActive:1}

Question

questions

Q&A forum questions (Stack Overflow-style).

FIELD	TYPE	CONSTRAINTS / NOTES
author *	ObjectId→User	indexed
university *	ObjectId→University	indexed
title *	String	max 150; text index
body	String	max 3000; text index
branch / subject / semester	String	filters; subject in text index
upvotes	[ObjectId→User]	voter list
upvoteCount	Number	default 0, indexed — denormalized sort key
answersCount	Number	default 0 — denormalized
acceptedAnswer	ObjectId→Answer	null until the author accepts one
isActive	Boolean	default true

Indexes: text {title, body, subject} · {university:1, isActive:1, createdAt:-1}

Answer

answers

Answers to forum questions.

FIELD	TYPE	CONSTRAINTS / NOTES
question *	ObjectId→Question	indexed
author *	ObjectId→User	indexed
university *	ObjectId→University	indexed
body *	String	max 3000
upvotes	[ObjectId→User]	voter list
upvoteCount	Number	default 0 — denormalized
isActive	Boolean	default true

Indexes: {question:1, isActive:1}

StudyGroup

studygroups

Student study groups with membership, external links and scheduled sessions.

FIELD	TYPE	CONSTRAINTS / NOTES
name *	String	text index
description	String	text index
subject *	String	text index
tags	[String]	indexed; also in text index
image	String	cover image URL
creator *	ObjectId→User	auto-added to members (pre-save hook)
members	[ObjectId→User]	deduplicated in pre-save
maxMembers	Number	default 10, min 2
links	Object	whatsapp / telegram / discord / googleMeet URLs
customLinks	[Subdoc]	{label, url}
nextSession	Object	{at: Date, mode: online offline, location, meetingLink}
university *	ObjectId→University	indexed
isActive	Boolean	default true, indexed
notifications.sessionReminderSent	Boolean	default false, indexed — one reminder per scheduled session

Indexes: text {name, description, subject, tags} · {university:1, isActive:1} · {university:1, isActive:1, nextSession.at:1}

Hooks & virtuals: pre-save ensures creator ∈ members and dedupes members · virtual memberCount

AcademicRecord

academicrecords

One document per user per semester — subjects with credits and grades, used for SGPA/CGPA.

FIELD	TYPE	CONSTRAINTS / NOTES
user *	ObjectId→User	indexed
university *	ObjectId→University	indexed
semester *	Number	1–12
subjects	[Subdoc]	{name*, credits* (0.5–30), grade* ∈ A+ A B+ B C+ C D F} (no _id)

Indexes: {user:1, semester:1} unique — one record per user per semester

AttendanceSubject

attendancesubjects

Per-subject attendance tracker ("can I bunk?" calculator).

FIELD	TYPE	CONSTRAINTS / NOTES
user *	ObjectId→User	indexed
university *	ObjectId→University	indexed
name *	String	subject name
attended	Number	default 0, min 0 — classes attended
held	Number	default 0, min 0 — classes held so far
target	Number	default 75 (1–100) — minimum % to stay above

Indexes: {user:1, createdAt:1}

TimetableEntry

timetableentries

Weekly class schedule slots; drives class reminders.

FIELD	TYPE	CONSTRAINTS / NOTES
user *	ObjectId→User	indexed
university *	ObjectId→University	indexed
subject *	String	max 80
dayOfWeek *	Number	0 (Sunday) – 6 (Saturday), JS Date convention
startTime * / endTime *	String	24h "HH:mm" (regex-validated), local campus time (IST)
room / professor	String	max 60 each
lastReminderDay	String	"YYYY-MM-DD" of the last reminder sent — prevents duplicates

Indexes: {user:1, dayOfWeek:1, startTime:1}

5 · Campus Life

Confession

confessions

Anonymous confessions feed. Author is stored for moderation but never exposed via the API.

FIELD	TYPE	CONSTRAINTS / NOTES
author *	ObjectId→User	indexed; private — never serialized
university *	ObjectId→University	indexed
content *	String	max 1000
reactions	[Subdoc]	{user* → User, type* ∈ heart laugh wow sad}
reactionsCount	Number	default 0, indexed — denormalized for "Top" sort
comments	[Subdoc]	{author* → User (hidden), content* (max 500)} + timestamps
reports	[Subdoc]	{user* → User, reason (max 300), createdAt}
isHidden	Boolean	default false — auto-hidden once reports cross the threshold
isActive	Boolean	default true

Indexes: {university:1, isActive:1, isHidden:1, createdAt:-1}

Event

events

Campus events (fests, workshops, talks) with attendance and pre-start reminders.

FIELD	TYPE	CONSTRAINTS / NOTES
organizer *	ObjectId→User	indexed
university *	ObjectId→University	indexed
title *	String	max 120; text index
description	String	max 2000; text index
club	String	max 80 — organizing club/society; text index
category *	String enum	Cultural Technical Sports Workshop Talk Other
venue	String	max 120
startAt *	Date	indexed
endAt	Date	optional
attendees	[ObjectId→User]	RSVP list
reminderSent	Boolean	default false — one reminder ~1h before start
isActive	Boolean	default true

Indexes: text {title, description, club} · {university:1, isActive:1, startAt:1}

LostFoundItem

lostfounditems

Lost & found posts — reunite students with their stuff.

FIELD	TYPE	CONSTRAINTS / NOTES
title *	String	text index
description	String	text index
kind *	String enum	lost found — did the poster lose it or find it
category *	String enum	Electronics ID & Cards Books & Notes Clothing Keys Accessories Other
location	String	where lost/found
images	[String]	photo URLs
reporter *	ObjectId→User	
university *	ObjectId→University	indexed
status	String enum	open returned (default open, indexed)
isActive	Boolean	default true

Indexes: text {title, description} · {university:1, isActive:1}

Ride

rides

Ride-sharing posts — split cabs to the station/airport/home.

FIELD	TYPE	CONSTRAINTS / NOTES
poster *	ObjectId→User	indexed
university *	ObjectId→University	indexed
from * / to *	String	max 80 each
departureAt *	Date	indexed
seatsTotal *	Number	1–8 — seats for co-passengers (poster's own seat excluded)

FIELD	TYPE	CONSTRAINTS / NOTES
passengers	[ObjectId→User]	users who joined
note	String	max 300
isActive	Boolean	default true

Indexes: {university:1, isActive:1, departureAt:1} **Virtuels:** title = "from → to" (used by the shared conversation loader)

6 · Notifications, Devices & Moderation

Notification

notifications

In-app notification feed with a polymorphic pointer to the related entity.

FIELD	TYPE	CONSTRAINTS / NOTES
user *	ObjectId→User	indexed — the recipient
type *	String enum	message study-group marketplace system friend
title * / description *	String	display text
read	Boolean	default false
relatedId	ObjectId (refPath)	dynamic ref via <code>relatedModel</code>
relatedModel	String enum	Listing StudyGroup Message User Conversation

Indexes: {user:1, createdAt:-1} · {user:1, read:1}

DeviceToken

devicetokens

Push-notification registration tokens per device.

FIELD	TYPE	CONSTRAINTS / NOTES
user *	ObjectId→User	indexed
university *	ObjectId→University	indexed
token *	String	unique — the push token itself
platform	String enum	web android ios (default web)

Report

reports

User reports on any content type; feeds the admin moderation queue.

FIELD	TYPE	CONSTRAINTS / NOTES
reporter *	ObjectId→User	indexed
university *	ObjectId→University	indexed
targetType *	String enum	listing document confession question answer lostfound ride event
targetId *	ObjectId	indexed — id of the reported content (polymorphic)
reason *	String	max 300
snapshot	Object	{title, content (max 500)} — copy of the reported content so the queue stays reviewable after deletion
status	String enum	open resolved dismissed (default open, indexed)
handledBy / handledAt	ObjectId→User / Date	which admin resolved it, and when

Indexes: {university:1, status:1, createdAt:-1} · {reporter:1, targetType:1, targetId:1} unique — one report per user per item

7 · Relationship Map

University is the tenant root — referenced (required + indexed) by every collection except Notification.

User is referenced by: Listing.seller, Offer.buyer/seller, Conversation.participants, Message.sender, Document.uploader/upvotes/comments.author, Question.author/upvotes, Answer.author/upvotes, StudyGroup.creator/members, AcademicRecord.user, AttendanceSubject.user, TimetableEntry.user, Confession.author/reactions/comments/reports, Event.organizer/attendees, LostFoundItem.reporter, Ride.poster/passengers, Friendship.requester/recipient/users, Notification.user, DeviceToken.user, Report.reporter/handledBy, and User.savedListings → Listing.

Polymorphic links: Conversation.listingId points at a Listing, LostFoundItem or Ride depending on contextType ; Notification.relatedId uses refPath (Listing/StudyGroup/Message/User/Conversation); Report.targetId is typed by targetType (8 content types).

Denormalized counters (kept in sync by controllers): Question.upvoteCount/answersCount, Answer.upvoteCount, Document.upvoteCount/downloadCount, Confession.reactionsCount, Conversation.lastMessage/lastMessageAt, Offer.seller.